BiLSTM + Bahdanau Attention for Named Entity Recognition

1 Architecture Description

1.1 Model Overview

The proposed architecture is a Bidirectional Long Short-Term Memory (BiLSTM) network augmented with a Bahdanau Attention mechanism for Named Entity Recognition (NER).

Encoder: A Bidirectional LSTM processes each input sequence in both forward and backward directions. This allows the model to capture both past and future contextual dependencies for every token.

Bahdanau Attention: The attention mechanism computes alignment scores between each hidden state and all other hidden states in the sequence. These scores are normalized using a softmax function to obtain attention weights. A context vector is then computed as a weighted sum of hidden states:

$$e_{t,i} = v^T \tanh(W_h h_i + W_s h_t)$$

$$\alpha_{t,i} = \frac{\exp(e_{t,i})}{\sum_j \exp(e_{t,j})}$$

$$c_t = \sum_i \alpha_{t,i} h_i$$

The context vector c_t is concatenated with the current hidden state h_t.

Classification Layer: The concatenated representation $[h_t; c_t]$ is passed through a fully connected layer to predict BIO tags.

1.2 Preprocessing

- Tokenization performed via JSONL parsing.
- Vocabulary built from train and validation sets.
- Special tokens: `<PAD>` and `<UNK>`.
- Sequences padded/truncated to maximum length 128.
- Pre-trained embeddings loaded into embedding layer.

1.3 Embedding Choice

GloVe (100d) embeddings were selected as the primary embedding type after ablation experiments.

2 Training Setup and Hyperparameters

The model was hyper-optimized under a strict 15-minute training window constraint.

Hyperparameter	**Value**
Hidden Dimension	128
Embedding Type	GloVe (100d)
Batch Size	64
Max Sequence Length	128
Learning Rate	0.003
Epochs	12 (per layer setting)
Optimizer	Adam
Layers (L)	$\{1, 2, 3\}$

Table 1: Training Hyperparameters

3 Training and Validation Curves

3.1 Loss Curves for L = 1

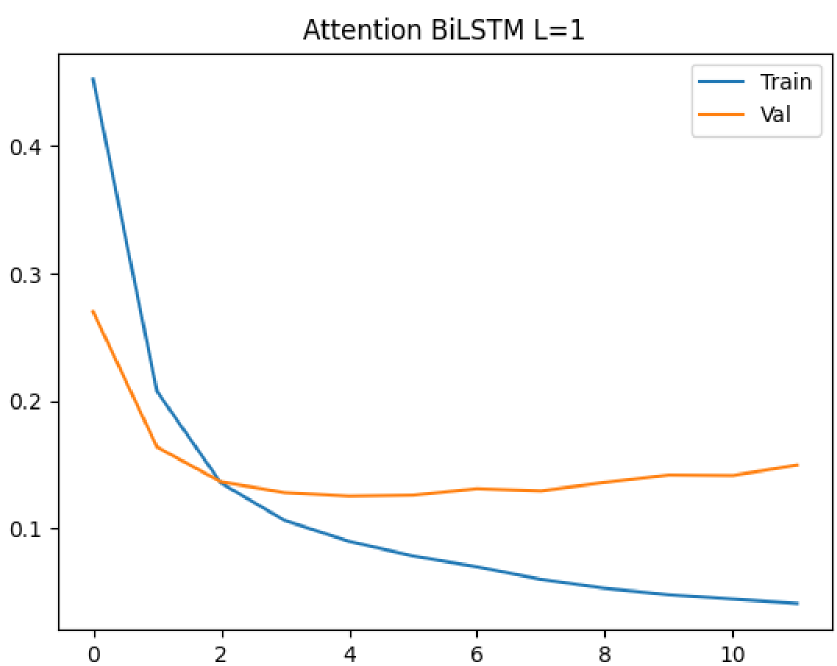

Figure 1: Training and Validation Loss for L=1

3.2 Loss Curves for L = 2

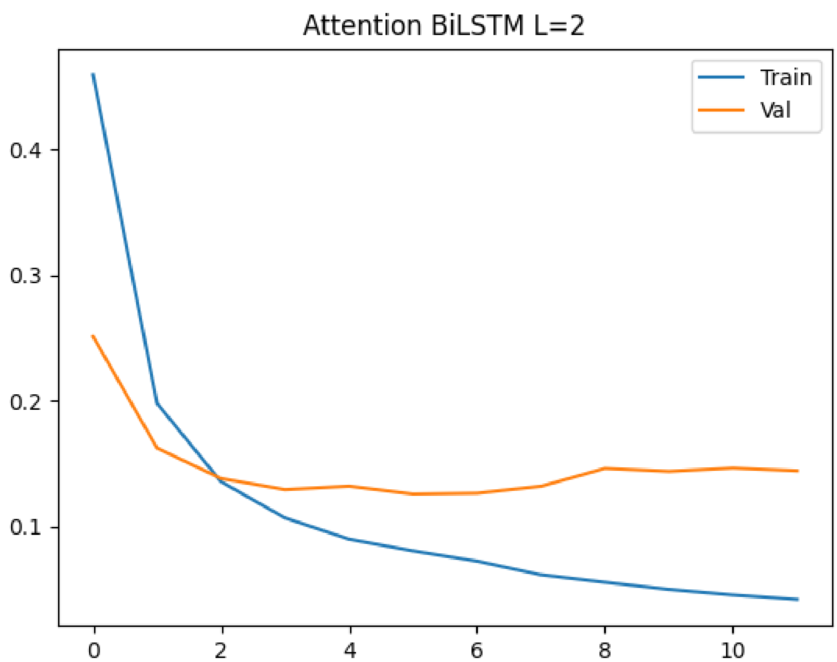

Figure 2: Training and Validation Loss for L=2

3.3 Loss Curves for L = 3

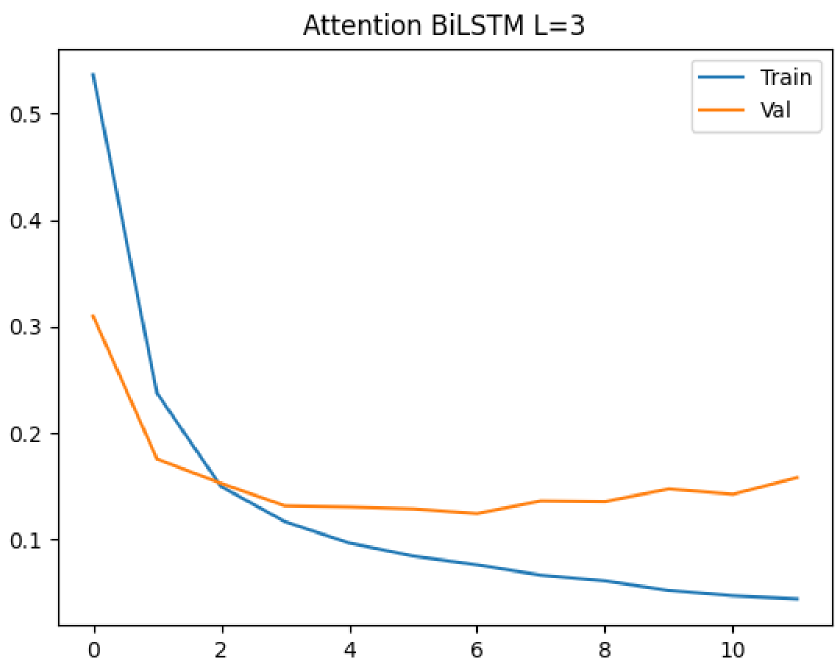

Figure 3: Training and Validation Loss for L=3

4 Results

4.1 Complete Results Table

Layers (L)	**FreeMatch-F1**	**Strict EM**
1 (Champion)	0.7273	0.4934
2	0.7044	0.4671
3	0.6984	0.4714

Table 2: Final Performance Comparison Across Depths

4.2 Best Model Performance Screenshot

Best Performing Model Metrics (Q3)

Model Config	FreeMatch-F1	Strict EM
L=1 (GloVe 100d)	0.7273	0.4934

Figure 4: EM and FreeMatch-F1 Screenshot for Best Model (L=1)

5 In-Depth Analysis

5.1 Effect of Attention

The Bahdanau Attention mechanism improves the BiLSTM by allowing the model to dynamically focus on relevant tokens when predicting each label. In NER, entity classification heavily depends on context (e.g., distinguishing `B-ORG` from `B-LOC`). Attention enables better long-range dependency modeling compared to vanilla BiLSTM.

5.2 Effect of Increasing Depth

Increasing depth from $L = 1$ to $L = 2$ initially provided performance improvements due to richer hierarchical feature extraction. However, at $L = 3$, performance slightly decreased in FreeMatch-F1 and did not improve Strict EM.

Possible reasons:

- Over-parameterization relative to dataset size.
- Increased training difficulty within limited time budget.
- Overfitting, as evidenced by widening train-validation loss gap.

5.3 FreeMatch-F1 vs Strict EM

FreeMatch-F1 remains consistently higher than Strict EM across all configurations.

Reason:

Strict Exact Match requires every token in a sentence to be predicted correctly. A single token error results in zero score for the entire sequence.

FreeMatch-F1 is token-level and rewards partial correctness. Thus:

- A sentence that is mostly correct still contributes positively to F1.

- The model may learn good token-level predictions without perfect sequence-level consistency.

As depth increases, the model may improve local predictions (maintaining F1), but sequence-level consistency may not improve proportionally, explaining the divergence between metrics.

6 Conclusion

The BiLSTM + Bahdanau Attention architecture effectively models contextual dependencies for NER. The optimal configuration under the 15-minute constraint was $L = 1$ with GloVe embeddings, achieving:

$$\text{FreeMatch-F1} = 0.7273$$

$$\text{Strict EM} = 0.4934$$

Increasing depth beyond one layer resulted in diminishing returns and slight generalization degradation.